

CENTRAL CALIFORNIA
EMERGENCY MEDICAL SERVICES
A Division of the Fresno County Department of Public Health

Manual	Emergency Medical Services Administrative Policies and Procedures	Policy Number 530.04
Subject	Paramedic Treatment Protocols CARDIAC ARREST – MEDICAL ADULT	Page 1 of 3
References	Title 22, Division 9, Chapter 3.3 of the California Code of Regulations	Effective Fresno County: 01/15/82 Kings County: 04/10/89 Madera County: 06/15/85 Tulare County: 04/19/05

STANDING ORDERS			
Assess need for CPR	Assess for carotid or femoral pulse for <u>no longer than 10 seconds</u> . If in doubt, begin chest compressions immediately.		
Begin Compressions	Perform high quality manual chest compressions (see special considerations)		
Assess Rhythm	Apply cardiac monitor and pads. Treat according to rhythm.		
Defibrillation	If VF/PVT, defibrillate as soon as possible after compressions are initiated. First defibrillation attempt should be at 200J.		
Ventilate and Oxygenate	Provide ventilations with bag valve mask and high flow O ₂ to achieve chest rise. Do not over-ventilate.		
Advanced Airway	Secure with supraglottic or ET tube. Supraglottic airway is preferred for initial airway management. Continuous ETCO ₂ waveform capnography is mandatory.		
IV/IO access	IV is preferred. IO may be used at provider discretion.		
PEA/Asystole		Ventricular Fibrillation/Ventricular Tachycardia	
Epinephrine	1 mg (1:10,000) IV/IO. Repeat every 3-5 minutes. Max total dose 5 mg. Contact base for additional doses.	Defibrillation	Every 2 minutes while VF/VT persists. First defibrillation attempt should be at 200J, subsequent attempts should be maximum energy (200J for Zoll, 360J for Lifepak).
Lactated Ringers	1000 ml bolus IV/IO if suspected hypovolemia	Epinephrine	After defibrillation x 2: 1 mg (1:10,000) IV/IO. Repeat every 3-5 minutes. Max total dose 5 mg. Contact base for additional doses.

STANDING ORDERS CONTINUED ON NEXT PAGE

Approved By	Daniel J. Lynch (Signature on File at EMS Agency)	Revision
EMS Division Manager		06/01/2025
EMS Medical Director	Miranda Lewis, MD (Signature on File at EMS Agency)	

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STANDING ORDERS	
PEA/Asystole	Ventricular Fibrillation/Ventricular Tachycardia
	Amiodarone After defibrillation x 3 (for refractory or recurrent VF/VT without pulses): Amiodarone 300 mg IV/IO Repeat Amiodarone 150 mg IV/IO as needed one time after defibrillation x 2. Max total dose 450 mg Magnesium 2 grams IV/IO push. ONLY for suspected Torsade de Pointes

Calcium Chloride	1000 mg IV/IO once. Administer over 2 minutes. To be administered ONLY in dialysis patients with suspected hyperkalemia. Consider early administration.
Sodium Bicarbonate	50 mEq/50ml IV/IO once. To be administered ONLY in suspected sodium channel blocker/tricyclic overdose or in dialysis patients with suspected hyperkalemia.
Return of Spontaneous Circulation/Post Arrest Care	
Assess vitals and monitor	Recheck BP every 3-5 minutes at minimum. Place on cardiac monitor and continuous EtCO ₂ waveform capnography.
Ventilate and Oxygenate	High flow O ₂ . Low volume ventilations at 10-12 breaths per minute. Do not overventilate or hyperventilate. Target ETCO ₂ to 35-45 mmHg.
If SBP < 90 mmHg	Fluid bolus: 250-1000 ml Lactated Ringers or Normal Saline IV/IO wide open to maintain BP > 90 mmHg. Do not administer if there is suspected volume overload or severe heart failure. Epinephrine drip: - If unresponsive to fluid, exam suggests volume overload, or concern for impending rearrest due to hypotension. 1 mg Epinephrine in 250 ml NS. Administer <u>only</u> with microdrip (60 gtt/ml) tubing. Start at 2 mcg/min (1 drop every 2 seconds) and titrate carefully at 1 mcg/min increments to target BP >90 mmHg. See drip chart in special considerations. Max dose 8 mcg/min (2 drops per second).
Check blood glucose	If <60 mg/dL: - Dextrose 10% 250 mL IV/IO preferred. Dextrose 50% 50 mL may be used if D10 unavailable. - If unable to obtain IV/IO access, Glucagon 1mg (1 ml) IM
12 lead ECG	If *** STEMI, transmit ECG and alert STEMI receiving center.
If patient rearrests	Immediately resume compressions and treat according to rhythm as above.
Transport	
Transport with ROSC	Patients with ROSC should be transported to the nearest STEMI receiving center if transport time is less than 30 minutes per EMS Policy #547. If available, place LUCAS device on patient on standby in case of rearrest. Begin transport once initial stabilizing measures in place.

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Transport with ongoing CPR	If patient does not meet termination of resuscitation criteria (per EMS Policy #549) and no ROSC <u>after 20 minutes of resuscitation</u> , begin transport with ongoing CPR to nearest hospital per Policy #547.
Contact Hospital	Per EMS Policy #530.02

BASE HOSPITAL ORDERS

*Needle Thoracostomy In suspected tension pneumothorax. Refer to EMS Policy #530.02

SPECIAL CONSIDERATIONS AND PRIORITIES

- The treatment goals of the medical cardiac arrest patient are to increase odds of ROSC prior to transport and stabilize patients on scene to prevent re-arrest. Transport decision should be made after 20 minutes of on scene resuscitation unless special circumstances exist warranting early transport (environmental hypothermia, pregnancy >20 weeks gestation, drowning) or if there are scene safety issues.
- High Performance CPR includes the following elements:
 - Rate 110 per minute (use metronome to guide rate)
 - Depth 2 inches
 - Allow full chest recoil
 - Minimize interruptions in chest compressions (goal >90% compression fraction)
 - Avoid overventilation. Deliver only enough ventilatory volume to achieve chest rise.
- Anterolateral pad placement is preferred initially to minimize interruptions in compressions. The anterior pad should be placed just below the right clavicle. The lateral pad should be placed in the left lower chest centered in the midaxillary line.
- If three shocks have been delivered and patient remains in a shockable rhythm, consider changing pad placement to anteroposterior position. This should be done in a choreographed, stepwise fashion to minimize interruptions in compressions.
- Hypothermia or drowning- Contact the Base Hospital as soon as possible. Transport should be initiated after initial steps of resuscitation. Continue resuscitation until rewarming has been implemented in the Emergency Department.
- Considerations in pregnant patients > 20 weeks gestation:
 - If personnel are available, one individual should manually displace the uterus to the left during compressions to relieve the pressure of the gravid uterus on the inferior vena cava.
 - IV/IO should be placed above the diaphragm
 - Prioritize oxygenation, consider early SGA placement.
 - Prepare for early transport after 5 minutes CPR.
- Sudden decrease in ETCO₂ is suggestive of rearrest. Immediately recheck a carotid or femoral pulse and prepare to resume compressions. Target ETCO₂ is 35-45 mmHg. ETCO₂ may be elevated immediately after ROSC but should normalize. Do not hyperventilate.
- Epinephrine drip should be reserved for patients who are unresponsive to fluid challenge or have evidence of significant volume overload or heart failure with hypotension. Epinephrine is arrhythmogenic in high doses and must be titrated carefully. DO NOT run wide open.

EPINEPHRINE DRIP CHART	
Mix 1 mg (1 ml) 1:1,000 Epinephrine in 250 ml of Normal Saline	
Use only with microdrip tubing (calibration of 60 drops per 1 ml)	
1 mcg drip = 15 gtt/min	5 mcg drip = 75 gtt/min
2 mcg drip = 30 gtt/min	6 mcg drip = 90 gtt/min
3 mcg drip = 45 gtt/min	7 mcg drip = 105 gtt/min
4 mcg drip = 60 gtt/min	8 mcg drip = 120 gtt/min